A Decomposition Theorem for Canonical Processes with Log-Concave Tails

Hanchao Wang

Shandong University

Joint work with Xuanang Hu and Xinglong Wu.

Roadmap

1. The Gaussian benchmark
2. The Bernoulli decomposition
3. Log-concave tails without Δ_2
4. Main techniques

Michel Talagrand

2024 Abel Prize

Awarded by the Norwegian Academy of Science and Letters.

- **Suprema of stochastic processes**
- **Concentration of measure**
- **Spin glasses**



Michel Talagrand receives the 2024 Abel Prize at the University Aula, 21 May 2024.

Canonical Gaussian processes: one metric

Let $g_1, \ldots, g_n$ be independent standard Gaussian variables and let $T \subset \mathbb{R}^n$ be finite and nonempty. Set

$$G_t = \sum_{i=1}^{n} t_i g_i, \qquad S_G(T) = \mathsf{E} \max_{t \in T} G_t.$$

The canonical metric is

$$d(s,t) = \left(\mathsf{E}|G_s - G_t|^2\right)^{1/2} = \|s - t\|_2.$$

Sudakov minoration

If $d(t_j, t_k) \geq a$ for $j \neq k$, then, for $m \geq 2$,

$$\mathsf{E} \max_{j \leq m} G_{t_j} \gtrsim a\sqrt{\log m}.$$

Admissible partitions and the Gaussian γ_2 functional

A nested sequence of partitions $(\mathcal{A}_k)_{k\geq 0}$ of T is **admissible** if

$$\mathcal{A}_0 = \{T\}, \qquad |\mathcal{A}_k| \leq 2^{2^k} \quad (k \geq 1).$$

Let $A_k(t)$ denote the cell containing t. Define

$$\gamma_2(T,d) = \inf_{(\mathcal{A}_k)} \max_{t\in T} \sum_{k\geq 0} 2^{k/2} \operatorname{diam}_d A_k(t).$$

Talagrand's majorizing-measure theorem

For a centered Gaussian process,

$$S_G(T) \asymp \gamma_2(T,d).$$

The constants are universal.

The majorizing-measure formulation

For a probability measure μ on T, let

$$I_\mu(t) = \int_0^\infty \sqrt{\log \frac{1}{\mu(B_d(t,\epsilon))}}\, d\epsilon.$$

Then

$$S_G(T) \asymp \inf_{\mu \in \mathcal{P}(T)} \max_{t \in T} I_\mu(t).$$

expected supremum ◂▸ admissible partitions ◂▸ majorizing measures

Gaussian lesson: one metric encodes all relevant scales.

Why Gaussian chaining is no longer exact

Let (ε_i) be independent symmetric signs. Write

$$B_t = \sum_{i=1}^{n} t_i \varepsilon_i, \qquad S_B(T) = \mathsf{E} \max_{t \in T} B_t.$$

For $T = \{e_1, \ldots, e_m\} \subset \mathbb{R}^m$, $m \geq 2$,

$$S_B(T) = \mathsf{E} \max_{j \leq m} \varepsilon_j \leq 1, \qquad \|e_j - e_k\|_2 = \sqrt{2} \quad (j \neq k).$$

In contrast,

$$\gamma_2(T, \|\cdot\|_2) \asymp \sqrt{\log m}.$$

The obstruction

Boundedness can come from the bounded coordinates themselves, rather than from Gaussian-type cancellation.

The Bednorz–Latała decomposition theorem

Bednorz–Latała (2014)

For every finite $T \subset \mathbb{R}^n$,

$$S_B(T) \asymp \inf_{T \subset T_1 + T_2} \left\{ \sup_{a \in T_1} \|a\|_1 + \gamma_2(T_2, \|\cdot\|_2) \right\}.$$

The infimum is over finite sets in the ambient space.

The first part: no cancellation

$$\left| \sum_i a_i \varepsilon_i \right| \leq \|a\|_1.$$

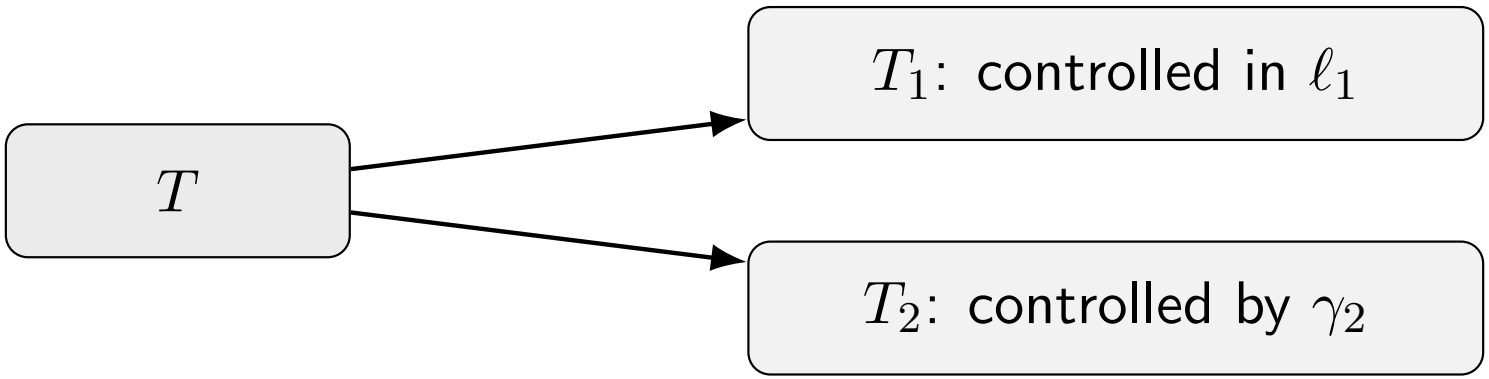
A deterministic ℓ_1 bound suffices.

The second part: chaining

$$S_B(T_2) \lesssim \gamma_2(T_2, \|\cdot\|_2).$$

The increments admit Gaussian-type control.

What the decomposition changes



The decomposition is **extrinsic**: T_1 and T_2 need not lie in T.

Talagrand's partitioning method and chopping maps make it possible to separate these two mechanisms.

The next question

What replaces the ℓ_1 part when the coordinate magnitudes are random and have arbitrary log-concave tails?

The canonical process with log-concave tails

Let $Y_1, \ldots, Y_n$ be independent symmetric random variables. Set

$$N_i(u) = -\log \mathsf{P}(|Y_i| \geq u), \qquad q_i(p) = \sup\{u \geq 0 : N_i(u) \leq p\}.$$

The assumption is that N_i is convex. Then q_i is nondecreasing and concave. We normalize

$$q_i(1) = 1, \qquad Y_t = \sum_{i=1}^{n} t_i Y_i, \qquad S_Y(T) = \mathsf{E} \max_{t \in T} Y_t.$$

Quantile representation

For independent mean-one exponentials (E_i) and independent signs (ε_i),

$$(Y_i)_i \stackrel{\mathrm{d}}{=} \big(\varepsilon_i q_i(E_i)\big)_i.$$

Thus q_i describes the random magnitude in coordinate i.

The moment-chaining functional

The original process determines a family of distances:

$$d_p(s,t) = \|Y_s - Y_t\|_p, \qquad p \geq 1.$$

For an admissible sequence $(\mathcal{A}_k)$, let $A_k(t)$ be the cell containing t. Define

$$\gamma_Y(T) = \inf_{(\mathcal{A}_k)} \max_{t\in T} \sum_{k\geq 0} \operatorname{diam}_{d_{2^k}} A_k(t).$$

The upper bound is general

Chaining gives $S_Y(T) \leq C\gamma_Y(T)$. The difficult question is whether the reverse estimate holds.

Unlike the Gaussian case, the geometry may change with the moment order p.

Classical result I: Latała (1997)

Assume the tail functions $N_i : [0,\infty) \to [0,\infty)$ are finite-valued and convex, and that, uniformly in i,

$$N_i(2u) \le K N_i(u), \qquad u > 0.$$

Two-sided characterization under Δ_2

For independent symmetric coordinates with these tails,

$$c_K \gamma_Y(T) \le S_Y(T) \le C \gamma_Y(T).$$

The constants are independent of the dimension and the index set.

This is the moment-chain formulation of Latała's geometric characterization of boundedness.

Classical result II: Latała–Tkocz (2015)

After variance normalization, suppose the independent symmetric coordinates satisfy, uniformly in i,

$$\|Y_i\|_p \le \alpha \frac{p}{q} \|Y_i\|_q \quad (p \ge q \ge 2), \qquad \|Y_i\|_{Lp} \ge 2\|Y_i\|_p \quad (p \ge 2),$$

where $\alpha \ge 1$ and $1 < L < \infty$.

Regular moment growth and positive speed

$$c_{\alpha,L}\gamma_Y(T) \le S_Y(T) \le C\gamma_Y(T).$$

This extends the log-concave-tail theorem to a larger class of coordinates.

For log-concave tails, the first condition holds with a universal α; Δ_2 supplies the second with $L = L(K)$.

In the finite-valued Δ_2 regime, $N_i(q_i(p)) = p$ for $p > 0$. Hence

$$N_i\big(2q_i(p)\big) \le Kp \quad \Longrightarrow \quad q_i(Kp) \ge 2q_i(p).$$

Iteration gives $q_i(K^j p) \ge 2^j q_i(p)$. Since $\|Y_i\|_p \asymp q_i(p)$ for $p \ge 2$, a fixed number of iterations yields

$$\|Y_i\|_{Lp} \ge 2\|Y_i\|_p, \qquad L = L(K).$$

For $N(u) = u^\eta$, $\eta \ge 1$, one has $q(p) = p^{1/\eta}$ and $q(2^\eta p) = 2q(p)$.

The increment geometry under Δ_2

With $q_i(1) = 1$, use the quadratic modification

$$\widehat{V}_i(u) = \begin{cases} u^2, & |u| \le 1, \\ 2N_i(|u|) - 1, & |u| > 1, \end{cases} \qquad C_p = \left\{ z : \sum_i \widehat{V}_i(z_i) \le p \right\}.$$

Let $\rho_p(h) = \sup_{z \in C_p} \langle h, z \rangle$. The moment estimate gives $\rho_p(h) \asymp \|\langle h, Y \rangle\|_p$ for $p \ge 2$.

A fixed dilation across moment levels

For $K_0 = 4K$,

$$\widehat{V}_i(2u) \le K_0 \widehat{V}_i(u) \implies 2C_p \subset C_{K_0 p} \implies \rho_{K_0 p}(h) \ge 2\rho_p(h).$$

Thus the same uniform scale growth holds for linear increments, not just for individual coordinates.

Why the original scales can be chained

The dilation $\rho_{K_0 p}(h) \geq 2\rho_p(h)$ has two consequences.

Earlier levels are summable for a fixed increment

For fixed h, $p \geq 2$ and $m \geq 0$,

$$\sum_{j=0}^{m} \rho_{K_0^j p}(h) \leq \rho_{K_0^m p}(h) \sum_{j=0}^{m} 2^{-(m-j)} \leq 2\rho_{K_0^m p}(h).$$

A high-moment ball is small at lower moment levels

$$\rho_{K_0^m p}(h) \leq a \quad \Longrightarrow \quad \rho_p(h) \leq 2^{-m} a.$$

These scale relations support the original-space partition construction, together with Sudakov minoration and concentration.

For independent symmetric signs,

$$\|\varepsilon_i\|_p = 1, \qquad \|\varepsilon_i - \varepsilon_j\|_p = 2^{1-1/p} \in [1,2) \quad (i \neq j,\ p \geq 1).$$

Increasing the moment order need not increase the distance by any fixed factor.

The coordinate-vector test

For $T_M = \{e_1, \ldots, e_M\}$,

$$S_B(T_M) = \mathsf{E} \max_{i \leq M} \varepsilon_i = 1 - 2^{1-M} \leq 1.$$

Nevertheless, the original moment-chain functional $\gamma_B(T_M)$ grows without bound as $M \to \infty$.

The difficulty is not the size of one coefficient: it is repeated accumulation over nearly unchanged moment scales.

Why the ordinary moment chain overcounts

Fix $k \geq 0$ and $M > 2^{2^k}$. Every admissible sequence satisfies

$$|\mathcal{A}_k| \leq 2^{2^k} < M.$$

Some cell $A \in \mathcal{A}_k$ contains two points. Choose $t \in A$. By nesting, $A_n(t)$ is not a singleton for $0 \leq n \leq k$, and therefore

$$\operatorname{diam}_{d_{2^n}} A_n(t) \geq 1 \qquad (0 \leq n \leq k).$$

Hence, for every admissible sequence,

$$\max_{t \in T_M} \sum_{n \geq 0} \operatorname{diam}_{d_{2^n}} A_n(t) \geq k+1, \qquad \gamma_B(T_M) \geq k+1.$$

The same bounded-coordinate contribution is charged at many levels, even though $S_B(T_M) \leq 1$.

Single-scale Sudakov does not require Δ_2

Latała's Sudakov minoration

For independent symmetric log-concave-tailed coordinates, let $p \geq 2$ and $m \geq e^p$. If

$$\|Y_{t_j} - Y_{t_l}\|_p \geq a \qquad (j \neq l),$$

then

$$\mathsf{E} \max_{j \leq m} Y_{t_j} \geq ca.$$

No Δ_2 condition is needed for this statement.

One scale versus all scales. The extra regularity is used to turn scale-by-scale lower bounds into the full original moment-chain estimate, without repeated accumulation.

Direct chaining versus decomposition

Under Δ_2

Original moment scales have uniform growth.
The partition method can give

$$S_Y(T) \asymp_K \gamma_Y(T)$$

directly in the original space.

Without Δ_2

Moment scales can become unreliable at high values.

Our argument separates an absolute value controlled part from a part suitable for chaining. Chopping is used first for capped exponentials.

Different conclusions: γ_Y is the original moment-chain functional. It is not the exponential chaining functional used in our decomposition below.

The exponential chaining functional

Let $S \subset \mathbb{R}^n$ be finite. Choose an admissible sequence $(\mathcal{A}_k)_{0\le k\le N}$ ending in singletons and vectors $u_k(A) \in \mathbb{R}^n$, constant on each cell, with

$$u_N(\{s\}) = s, \qquad u_k(s) = u_k(A_k(s)).$$

Define

$$\mathcal{C}_{\exp}(S) = \inf \max_{s\in S} \sum_{k=1}^{N} \left(2^{k/2} \|u_k(s) - u_{k-1}(s)\|_2 + 2^k \|u_k(s) - u_{k-1}(s)\|_\infty \right).$$

Why these two terms?

Under $q_i(1) = 1$, the increment estimate is

$$\|\langle h, Y\rangle\|_p \lesssim \sqrt{p}\, \|h\|_2 + p\|h\|_\infty.$$

At level k, admissibility calls for p of order 2^k.

Main result: a decomposition for arbitrary log-concave tails

Hu–Wang–Wu: the decomposition theorem

For every finite nonempty $T \subset \mathbb{R}^n$,

$$S_Y(T) \asymp \inf_{\substack{t_0 \in T \\ T - t_0 \subset T_1 + T_2}} \left\{ \mathsf{E} \max_{a \in T_1} \sum_i |a_i| \, |Y_i| + \mathcal{C}_{\exp}(T_2) \right\}.$$

Here T_1, T_2 are finite; the constants are universal.

Positive part

No cancellation is used:

$$\mathsf{E} \max_{a \in T_1} \sum_i |a_i| \, |Y_i|.$$

Chaining part

Small increments are summed:

$$\mathcal{C}_{\exp}(T_2).$$

Bernoulli and capped-exponential specializations

Bernoulli coordinates: $Y_i = \varepsilon_i$

The positive part is $\sup_{a\in T_1} \|a\|_1$. Thus

$$S_B(T) \asymp \inf_{t_0\in T,\ T-t_0\subset T_1+T_2} \left\{ \sup_{a\in T_1} \|a\|_1 + \mathcal{C}_{\exp}(T_2) \right\}.$$

This optimized expression is equivalent to the $\ell_1 + \gamma_2$ formula of Bednorz–Latała.

Capped exponentials: $\xi_i^{(\beta_i)} = \varepsilon_i \min(E_i, \beta_i)$

For $1 \le \beta_i \le \infty$ and $S_\beta(T) = \mathsf{E} \max_{t\in T} \langle t, \xi^{(\beta)} \rangle$,

$$S_\beta(T) \asymp \inf_{t_0\in T,\ T-t_0\subset T_1+T_2} \left\{ \sup_{a\in T_1} \sum_i \beta_i |a_i| + \mathcal{C}_{\exp}(T_2) \right\}.$$

Tail functions for the partition characterization

Set $b_i = \lim_{p\to\infty} q_i(p) \in [1,\infty]$ and

$$V_i(u) = \begin{cases} u^2, & |u| \le 1, \\ 2N_i(|u|) - 1, & 1 < |u| \le b_i, \\ +\infty, & |u| > b_i. \end{cases}$$

Let $\sigma_i(u) = V'_{i,+}(u)$ for $0 < u < b_i$.

The tail-slope functions

$$\Psi_i(a) = \int_0^{b_i} \sigma_i(u)\mathbf{1}_{\{\sigma_i(u)<a\}}\, du, \qquad a \ge 0,$$

$$\phi_j(s,t) = \sum_i \Psi_i\big(8^j|s_i - t_i|\big), \qquad j \in \mathbb{Z}.$$

In particular,

$$\Psi_i(a) = a^2/4 \quad (0 \le a \le 2), \qquad \Psi_i(a) \ge \min\{1, a^2/4\}.$$

The partition functional adapted to the tails

Choose nested partitions $(\mathcal{A}_k)$ with $\mathcal{A}_0 = \{T\}$ and $|\mathcal{A}_k| \le 2^{2^k}$, and scales $j_k(A) \in \mathbb{Z} \cup \{+\infty\}$ satisfying

$$\phi_{j_k(A)}(s,t) \le 2^k \qquad (s,t \in A \in \mathcal{A}_k),$$

$$\operatorname{diam}_{\|\cdot\|_2} T \le 8^{-j_0(T)}, \qquad j_k(A_k(t)) \le j_{k+1}(A_{k+1}(t)).$$

Definition

$$\Gamma_Y(T) = \inf_{(\mathcal{A}_k),(j_k)} \sup_{t\in T} \sum_{k\ge 0} 2^k 8^{-j_k(A_k(t))}.$$

The value $j_k(A) = +\infty$ is allowed only for a singleton, with $8^{-\infty} = 0$ and $\phi_\infty(t,t) = 0$.

Main result II: a partition characterization

Hu–Wang–Wu

For independent symmetric log-concave-tailed coordinates with $q_i(1) = 1$ and every nonempty $T \subset \ell_2$,

$$C^{-1}\Gamma_Y(T) \le S_Y(T) \le C\Gamma_Y(T).$$

The constant C is universal. No Δ_2 assumption is required.

For $t \in \ell_2$, the series $Y_t = \sum_{i\ge 1} t_i Y_i$ converges in L_2. For general T, set $S_Y(T) = \sup_{F\subset T,\, 0<|F|<\infty} S_Y(F)$.

Bernoulli coordinates

$$\Psi_i(a) = \min\{a^2/4, 1\}, \qquad \phi_j(s,t) = \sum_i \min\{8^{2j}|s_i - t_i|^2/4, 1\}.$$

This recovers the partition characterization for Bernoulli processes.

Capped exponentials

$$\xi_i^{(\beta_i)} = \varepsilon_i \min(E_i, \beta_i), \qquad 1 \le \beta_i \le \infty.$$

Define

$$V_\beta(u) = \begin{cases} u^2, & |u| \le 1, \\ 2|u| - 1, & 1 < |u| \le \beta, \\ +\infty, & |u| > \beta, \end{cases} \qquad C_p^\beta = \left\{ z : \sum_i V_{\beta_i}(z_i) \le p \right\}.$$

Its support function is

$$\rho_p^\beta(t) = \sup_{z \in C_p^\beta} \langle t, z \rangle, \qquad \rho_p^\beta(t) \lesssim \|\langle t, \xi^{(\beta)} \rangle\|_p \quad (p \ge 2).$$

If every $\beta_i = 1$, then $C_p^\beta = \sqrt{p}\, B_2^n \cap B_\infty^n$, the Bernoulli moment body; B_2^n, B_∞^n are the respective unit balls.

Why small coordinate amplitudes matter

For Bernoulli processes, suppose $m \geq 2$ and

$$\|t^j - t^l\|_2 \geq a \quad (j \neq l), \qquad \max_{j \leq m} \|t^j\|_\infty \leq b.$$

Talagrand's Euclidean Sudakov estimate is

$$\mathsf{E} \max_{j \leq m} \sum_i t_i^j \varepsilon_i \geq c \min \left\{ a\sqrt{\log m}, \frac{a^2}{b} \right\}.$$

To recover the Gaussian-sized lower bound

This estimate has Gaussian size when $b \lesssim a/\sqrt{\log m}$. Higher complexity therefore calls for smaller coordinate amplitudes.

Chopping acts on coefficients: it replaces a large coefficient by many bounded interval pieces. It does not turn the source into a Gaussian.

Talagrand's chopping maps

For $u < v$, define

$$\varphi_{u,v}(x) = \min\{v, \max\{x, u\}\} - \min\{v, \max\{0, u\}\}.$$

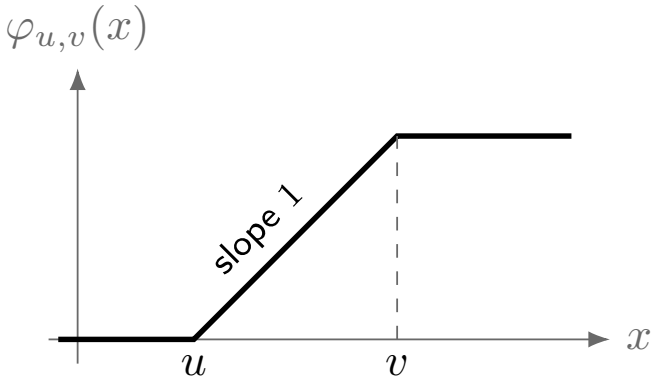


Illustration for $0 < u < v$.

Additivity across intervals

If $u_0 < \cdots < u_m$, then

$$\varphi_{u_0,u_m}(x) = \sum_{j=1}^{m} \varphi_{u_{j-1},u_j}(x).$$

Contraction of total variation

$$\sum_{j=1}^{m} |\varphi_{u_{j-1},u_j}(x) - \varphi_{u_{j-1},u_j}(y)| \le |x - y|.$$

What one coordinate becomes

With the unit intervals $[0,1]$, $[1,2]$, $[2,3]$,

$$2.7 = \varphi_{0,1}(2.7) + \varphi_{1,2}(2.7) + \varphi_{2,3}(2.7) = 1 + 1 + 0.7.$$

Thus the scalar 2.7 becomes the vector $(1, 1, 0.7)$.

Small pieces, with an exact reconstruction

If a finite interval grid covers $0, x, y$ and has mesh at most h, then

$$\sum_j \varphi_j(x) = x, \qquad \max_j |\varphi_j(x)| \le h,$$

$$\sum_j |\varphi_j(x) - \varphi_j(y)| = |x - y|.$$

Here $\varphi_j = \varphi_{u_{j-1},u_j}$ for consecutive grid points.

Independent layers: a comparison, not an identity

Using the same sign on all pieces changes nothing:

$$2.7\varepsilon = (1 + 1 + 0.7)\varepsilon.$$

The auxiliary process instead uses independent signs:

$$\varepsilon_1 + \varepsilon_2 + 0.7\varepsilon_3.$$

These random variables do not have the same distribution.

The probabilistic step

Give each interval piece its own independent copy of the original source. The resulting process has coefficients adapted to the current mesh.

Its usefulness comes from comparison inequalities for expected suprema, not from a pathwise identity with the original process.

Independent interval layers and refinement

Let G_i be a finite grid; $G_i^- = G_i \setminus \{\max G_i\}$, and u^+ is the successor of u. Set

$$\Phi_G(t) = \big(\varphi_{u,u^+}(t_i)\big)_{i,u\in G_i^-}, \qquad X_t(G) = \sum_i \sum_{u\in G_i^-} \varphi_{u,u^+}(t_i)\xi_{u,i}^{(\beta_i)},$$

with independent copies $\xi_{u,i}^{(\beta_i)}$. Write $F_G(A) = \mathsf{E}\max_{t\in A} X_t(G)$.

Refinement does not increase the expected supremum

If $G_i \subset G_i'$ with the same endpoints, then

$$F_{G'}(A) \le F_G(A).$$

Deleting chopped coordinates also cannot increase F_G.

Under a refinement into at most K pieces per interval, the support-function distances in the two enlarged spaces are comparable within a factor K.

Why one cannot chop arbitrarily finely

Take $x = Mh$ and split $[0, Mh]$ into M intervals of length h. In the enlarged space,

$$\Phi_h(x) - \Phi_h(0) = (h, \dots, h).$$

Consequently,

$$|x - 0| = Mh, \qquad \|\Phi_h(x) - \Phi_h(0)\|_2 = h\sqrt{M}, \qquad \|\Phi_h(x) - \Phi_h(0)\|_\infty = h.$$

A gain and a loss

Chopping reduces the maximum coordinate, but can also reduce the distance that is needed for the lower bound.

A smaller auxiliary process alone is not enough. The proof must also recover geometric information and control what is removed.

Recovering the original distance at the correct scale

For $s, t \in \mathbb{R}^n$, let G_h be the mesh-h grid between s_i and t_i in each coordinate, including both endpoints. Each interval inherits cap β_i.

$$\eta := \rho_p^{G_h}\big(\Phi_h(t) - \Phi_h(s)\big), \qquad p > 0.$$

The local inverse estimate

$$\eta \le \rho_p^{\beta}(t-s), \qquad \eta < \frac{ph}{2} \quad \Longrightarrow \quad \rho_p^{\beta}(t-s) \le 8\eta.$$

Here $\rho_p^{G_h}$ is the support-function distance on the enlarged coordinates.

Full mesh intervals are controlled by a weighted ℓ_1 estimate; at most two partial intervals remain in each original coordinate.

This recovers the original geometry provided that the chopped distance is small relative to the mesh scale ph.

Sudakov minoration plus concentration gives growth

Let $m \geq 8$, $p = \log m$, and suppose

$$\rho_p^\beta(t_j - t_l) \geq r \quad (j \neq l), \qquad \sup_{t \in B_j} \rho_p^\beta(t - t_j) \leq v,$$

where each B_j is finite and nonempty.

Growth inequality

$$S_\beta\left(\bigcup_{j=1}^m B_j\right) \geq \min_j S_\beta(B_j) + cr - Cv.$$

For $Z_j = \max_{t \in B_j} \langle t - t_j, \xi^{(\beta)} \rangle$,

$$\mathsf{E} \max_j \langle t_j, \xi^{(\beta)} \rangle \geq cr, \qquad \mathsf{E} \max_j |Z_j - \mathsf{E} Z_j| \leq Cv.$$

Separation supplies the gain; concentration controls the local fluctuations.

The greedy decomposition lemma

Given $m \geq 8$, $p = \log m$ and $v > 0$, a finite set A can be partitioned as

$$A = A_1 \cup \cdots \cup A_r, \qquad r \leq m.$$

Each cell satisfies one of the following alternatives.

Small diameter

$$\operatorname{diam}_{\rho_p^\beta} A_l \leq Kv.$$

The geometry is already controlled at the current scale.

A drop on every small subset

For nonempty $D \subset A_l$,

$$\operatorname{diam}_{\rho_p^\beta} D \leq v \implies S_\beta(D) \leq S_\beta(A) - v.$$

This alternative occurs for at most one cell.

Greedily remove enlarged balls around centers with maximal local expected supremum. The growth inequality handles the residual cell.

Separating large coordinates from small increments

Suppose $q \geq 1$ and $\rho_q^\beta(v) \leq \delta$. Set

$$h = \frac{2\delta}{q}, \qquad J = \{i : |v_i| > h\}.$$

Threshold estimate

$$\sum_{i\in J} \beta_i \leq \frac{q}{2}, \qquad \sum_{i\in J} \beta_i |v_i| \leq \delta, \qquad \|v\mathbf{1}_{J^c}\|_2 \leq \frac{2\delta}{\sqrt{q}}.$$

Large coordinates

Their contribution is paid for in the weighted ℓ_1 norm $\|x\|_{1,\beta} = \sum_i \beta_i |x_i|$.

Small coordinates

They retain the ℓ_2 and ℓ_∞ control needed for chaining.

Clipping supplies the two different controls

Let $u \in B$, $q \geq 1$, and $\sup_{t\in B} \rho_q^\beta(t-u) \leq \delta$. Set $h = 2\delta/q$. Choose $k_i \in \mathbb{Z}$ with $k_i h \leq u_i < (k_i+1)h$ and let

$$Q_i = [(k_i-1)h, (k_i+2)h], \qquad y_i(t) = P_{Q_i}(t_i).$$

The removed part

$$\sup_{t\in B} \|t - y(t)\|_{1,\beta} \leq \delta.$$

The retained part

$$\sup_{t\in B} \|y(t)-u\|_2 \leq \sqrt{12}\,\frac{\delta}{\sqrt{q}}, \qquad \sup_{t\in B} \|y(t)-u\|_\infty \leq \frac{4\delta}{q}.$$

The first estimate uses no cancellation; the other two are suited to chaining.

Clipping becomes selection of interval coordinates

Suppose a_i, b_i are grid points and $Q_i = [a_i, b_i]$. Additivity gives

$$\sum_{\substack{u \in G_i^- \\ [u,u^+] \subset Q_i}} \varphi_{u,u^+}(t_i) = \varphi_{a_i,b_i}(t_i) = P_{Q_i}(t_i) - P_{Q_i}(0).$$

Original coordinates versus interval coordinates

Clipping an original coordinate corresponds to retaining the interval pieces inside Q_i and deleting those outside it.

One can retain the small local variations of a coordinate without retaining the entire coordinate.

The identity is deterministic. After independent randomization of the pieces, the coordinate-deletion comparison controls the auxiliary expected supremum.

What the partition construction produces

Chopping, the greedy lemma and the coordinate-deletion lemma produce an admissible sequence $(\mathcal{A}_k)$, a map $z : T \to \mathbb{R}^n$ and representatives $u_k(A)$, with $u_N(\{t\}) = z(t)$, such that

$$\max_{t\in T}\|t - z(t)\|_{1,\beta} + \max_{t\in T}\sum_{k=1}^{N}\rho_{2^k}^{\beta}\big(u_k(t) - u_{k-1}(t)\big) \le CS_\beta(T).$$

How the costs are controlled

Refining the geometry reduces the radius; otherwise a local functional drops or coordinates are deleted. Along each branch, the functional drops telescope.

A final clipping of the increments gives $t - t_0 = a(t) + v(t)$ with

$$\max_{t\in T}\|a(t)\|_{1,\beta} + \mathcal{C}_{\exp}(v(T)) \le CS_\beta(T).$$

Dyadic tail slopes and capped coordinates

Work first with finite nonempty $T \subset \mathbb{R}^n$. For capped exponentials,

$$\psi_\beta(a) = \begin{cases} a^2/4, & 0 \le a \le 2, \\ 2\beta - 1, & a > 2. \end{cases}$$

For general tails, let $s_j = 2^{j+1}$, $c_j = 2^{-j}$, $j \ge 0$, and

$$A_{ij} = \{u \in (1, b_i) : s_j \le \sigma_i(u) < 2s_j\}, \qquad w_{ij} = s_j |A_{ij}|.$$

Whenever $w_{ij} \ge 1$, assign the cap $\beta_{ij} = (w_{ij} + 1)/2$, so $2\beta_{ij} - 1 = w_{ij}$. The expanded coefficient map is

$$(L_m t)_{i,*} = t_i, \qquad (L_m t)_{i,j} = c_j t_i \quad (0 \le j \le m,\ w_{ij} \ge 1).$$

Give $(i, *)$ cap 1 and (i, j) cap β_{ij}. Denote the independent capped variables by $\Xi^{(m)}$.

Theorem 1.2: the lower bound

Write

$$Z_i^{(m)} = \Xi_{i,*} + \sum_{\substack{0\le j\le m\\ w_{ij}\ge 1}} c_j \Xi_{i,j}, \qquad \langle L_m t, \Xi^{(m)}\rangle = \sum_i t_i Z_i^{(m)}.$$

Moment and convex-order comparisons

Uniformly in i, m, for $p \ge 1$,

$$\|Z_i^{(m)}\|_p \lesssim q_i(p), \qquad \mathsf{E} f(Z_i^{(m)}) \le \mathsf{E} f(CY_i)$$

for every convex f with finite expectations.

Let $\Gamma_{\Xi^{(m)}}$ be the partition functional for the expanded coordinates. The slope comparison, capped theorem and coordinatewise convexity yield

$$\begin{aligned}\Gamma_Y(T) &\lesssim \sup_m \Gamma_{\Xi^{(m)}}(L_m T)\\ &\lesssim \sup_m \mathsf{E}\max_{t\in T}\langle L_m t, \Xi^{(m)}\rangle \lesssim S_Y(T).\end{aligned}$$

Theorem 1.2: the upper bound and the decomposition

Choose cell representatives $\pi_k(t)$ and radii $u_k(t) = 8^{-j_k(A_k(t))}$ with

$$B = \max_{t\in T} \sum_{k\geq 0} 2^k u_k(t) \leq 2\Gamma_Y(T).$$

End at singletons, so $\pi_N(t) = t$, and let $t_0 = \pi_0(T)$. Clip each increment coordinate to $[-4u_{k-1}(t), 4u_{k-1}(t)]$:

$$h_{k,i}(t) = \mathrm{clip}_{4u_{k-1}(t)}\big(\pi_{k,i}(t) - \pi_{k-1,i}(t)\big).$$

The partition bounds imply

$$\|h_k(t)\|_\infty \leq 4u_{k-1}(t), \qquad \|h_k(t)\|_2 \leq 4\,2^{(k-1)/2}u_{k-1}(t).$$

For $v(t) = \sum_{k=1}^N h_k(t)$ and $a(t) = t - t_0 - v(t)$,

$$\mathcal{C}_{\exp}(v(T)) + \mathsf{E}\max_{t\in T}\sum_i |a_i(t)|\,|Y_i| \lesssim B \lesssim \Gamma_Y(T).$$

Tail-slope integration and $|Y_i| \overset{\mathrm{d}}{=} q_i(E_i)$ control the positive part. Thus $S_Y(T) \lesssim \Gamma_Y(T)$, with the decomposition $t - t_0 = a(t) + v(t)$.

Summary

Gaussian processes

One metric: the γ_2 functional exactly describes the expected supremum.

Bernoulli processes

Two mechanisms: an ℓ_1 part and a Gaussian-chaining part.

Independent symmetric log-concave tails

A partition characterization $S_Y(T) \asymp \Gamma_Y(T)$ and the positive-part-plus-chaining decomposition, without Δ_2.

Chopping maps $+$ **partitions** $+$ **tail slopes**

References: Gaussian and Bernoulli processes

[1] **X. Fernique (1975).** Régularité des trajectoires des fonctions aléatoires gaussiennes. *Lecture Notes in Mathematics* **480**, Springer, pp. 1–96.

[2] **M. Talagrand (1987).** Regularity of Gaussian processes. *Acta Mathematica* **159**, 99–149.

[3] **M. Talagrand (1996).** Majorizing measures: the generic chaining. *The Annals of Probability* **24**, 1049–1103.

[4] **W. Bednorz and R. Latała (2014).** On the boundedness of Bernoulli processes. *Annals of Mathematics* (2) **180**, 1167–1203.

References: log-concave tails and partition methods

[5] **E. D. Gluskin and S. Kwapień (1995).** Tail and moment estimates for sums of independent random variables with logarithmically concave tails. *Studia Mathematica* **114**, 303–309.

[6] **R. Latała (1997).** Sudakov minoration principle and supremum of some processes. *Geometric and Functional Analysis* **7**, 936–953.

[7] **R. Latała and T. Tkocz (2015).** A note on suprema of canonical processes based on random variables with regular moments. *Electronic Journal of Probability* **20**, no. 36, 1–17.

[8] **M. Talagrand (1994).** Constructions of majorizing measures, Bernoulli processes and cotype. *Geometric and Functional Analysis* **4**, 660–717.

References: decomposition theorems

[9] **M. Talagrand (2021).** *Upper and Lower Bounds for Stochastic Processes: Decomposition Theorems*. Second edition, Springer.

[10] **X. Hu, H. Wang and X. Wu.** The Decomposition Theorem for Canonical Processes with Log-Concave Tails. Manuscript.

Thank you!

Hanchao Wang

Shandong University

wanghanchao@sdu.edu.cn